By: M. KHALIL mth: 404 Lecture: 6-7

Rectilinear Motion:-

"The motion of a Particle along a stright line is called recti-linear motion"



Consider a particle which is moving along a stright line.
Suppose at a time 't' The Particle is at a point 'P' distance 'x' from The fixed point 'O' if 'v' and 'a' are The velocity and Acc of The particle at The time 't"

Then:

$$v = \frac{dx}{dt} \quad \text{——— (1)}$$

$$a = \frac{dv}{dt} \quad \text{——— (2)}$$

Putting The value of (1) in (2) we get

$$a = \frac{d}{dt}\left(\frac{dx}{dt}\right) = \frac{d^2x}{dt^2} \quad \text{——— (3)}$$

Since $a = \frac{dv}{dt} \Rightarrow a = \frac{dv}{dx} \cdot \frac{dx}{dt}$

$$a = v \cdot \frac{dv}{dx} \quad \text{——— (4)}$$

Eq (i—4) are called Diff equation of The rectilinear motion of The particle

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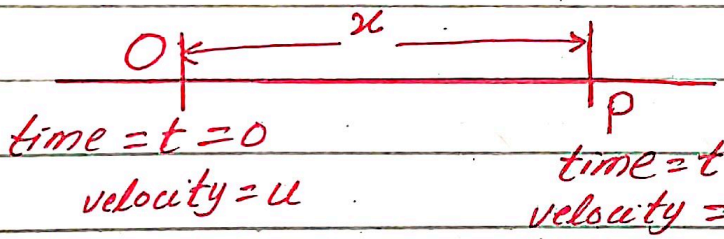
Lecture #6-7

Rectilinear Motion :-

"The motion of a body along a Stright line is called rectilinear motion :-

"Motion With Constant" Acc :-

Consider a Particle which is moving with Constant acceleration along a stright line.



Let at $t=0$ the particle is at point 'O' and is moving with velocity 'u'. After a time 't' the partile is at point 'P' at distance 'x' from 'O' and attains velocity 'v'.

We Know That $\Rightarrow \frac{dx}{dt} = v$

$$\frac{dx}{dt} = v$$

diff w.r. to 't'

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} \quad \because \frac{dv}{dt} = a \text{ (Acc)}$$

Integrating w.r.to. t

$$\frac{dx^2}{dt^2} = a \Rightarrow \int \frac{dx^2}{dt^2} = \int a$$

$$\int \frac{d}{dt}\left(\frac{dx}{dt}\right) = \int a \Rightarrow \int d\left(\frac{dx}{dt}\right) = \int a\,dt$$

$$\frac{dx}{dt} = at + A \quad \text{——(i)} \quad \because \frac{dx}{dt} = v$$

Initially when $t=0$,

velocity $= u$

Put in Eq (i) we get

$$u = a(0) + A \Rightarrow \boxed{A = u}$$

Equation (1) Becomes

$$v = at + u \quad \text{or} \quad v = u + at$$

Is the velocity of the particle at any time 't'

Again:-

$$\frac{dx}{dt} = u + at$$

$$dx = (u + at)\,dt$$

Integrate Both sides.

$$\int dx = \int (u+at)\,dt$$

$$\int dx = u\int dt + a\int t\,dt$$

$$x = ut + \frac{at^2}{2} + B \quad -②$$

Initialy when:- $t=0$, $x=0$

Put in Equ ② we get

$$0 = u(0) + a\left(\frac{0}{2}\right) + B$$

$$\boxed{B = 0}$$

So, Equation ② Beomes.

$$x = ut + \frac{1}{2}at^2$$

Is The distance Covered by The particle in Any time 't'

Question:- Drive The expression $v^2 - u^2 = 2ax$ where u and v are velocities of a particle at $t=0$ and time $= t$ a is The Acceleration and 'x' is The distance Covered in time 't'

Solution:- We Know That

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$v = u + at$ — (i) $x = ut + \frac{1}{2}at^2$ — (ii)

from ① $t = \frac{v-u}{a}$

using the value of 't' in ②

$$x = u\left(\frac{v-u}{a}\right) + \frac{1}{2}a\left(\frac{v-u}{a}\right)^2$$

$$2x = 2u\left(\frac{v-u}{a}\right) + a\left(\frac{v-u}{a}\right)^2$$

$$2x = \frac{2uv - 2u^2}{a} + \frac{a(v-u)^2}{a^2}$$

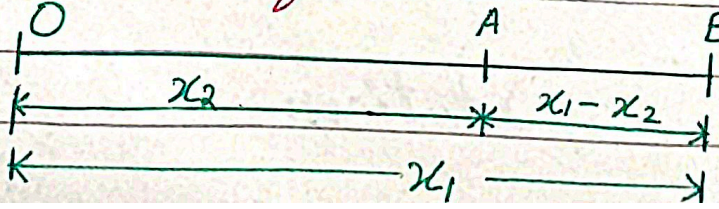
$$2ax = 2uv - 2u^2 + v^2 + u^2 - 2uv$$

$$2ax = v^2 - u^2$$

$$\boxed{2ax = v^2 - u^2}$$

Example:- Find the Distance traveled by the particle moving in a stright line with uniform aucleration in the nth unit of time:-

Solution:- Consider a Particle which is moving in a straight line with uniform ac,



Let x_1 and x_2 be the distance travelled by the particle in the first ('n') and (n-1) units of time resp.

Then By using the formulas:-

$$x = ut + \frac{1}{2}at^2$$

we have :-

$$x_1 = un + \frac{1}{2}an^2 \quad \text{——(i)}$$

$$x_2 = u(n-1) + \frac{1}{2}a(n-1)^2 \quad \text{——(ii)}$$

"Hence the distance travelled in nth unit of time is given By"

$$x_1 - x_2 = \left(un + \frac{1}{2}an^2\right) - \left(u(n-1) + \frac{1}{2}a(n-1)^2\right)$$

$$x_1 - x_2 = un + \frac{1}{2}an^2 - un + u - \frac{1}{2}a(n^2 - 2n + 1)$$

$$x_1 - x_2 = u + \frac{1}{2}a(n^2 - n^2 - 1 + 2n)$$

$$x_1 - x_2 = u + \frac{1}{2}a(2n - 1)$$

Example:- A Particle is moving in stright line starts from rest and its Acceleration uniform to attain a velocity of 60 miles per hour in 4 sec. Find the eq of motion and distance travelled by the partical in the last three seconds:?

Solution:- Given Data.

$t = 0$, $u = 0$, $v = 60 \text{ miles/h}$

$$v = \frac{60 \times 1760 \times 3}{3600} = 88 \text{ ft/sec}$$

$\because v = at + u \Rightarrow a = \frac{v-u}{t}$

$\therefore$ time $= 4$ sec

$$a = \frac{88 - 0}{4}$$

$$a = 22 \text{ ft/sec}^2$$

Distance in last 3rd sec $= ?$

$x =$ Distance in 4th sec $-$ Distance in 1st sec

$$x = \left[\frac{1}{2}(22)(4)^2 - 4(0)\right] - \left[\frac{1}{2}(22)(1)^2 + 0(1)\right]$$

$\because v = \frac{1}{2}at^2 + ut$

$$x = 176 \text{ ft} - 11 \text{ ft}$$

$$x = 165 \text{ ft}$$

Example:- A stone is let fall freely from height 100 feet. Find the time that it takes and the velocity that it acquires on reaching the ground.

Solution:- Given data

$h = 100 \text{ ft}$, $u = 0$, $a = g = 32 \text{ ft/sec}^2$

$v = ?$, $t = ?$

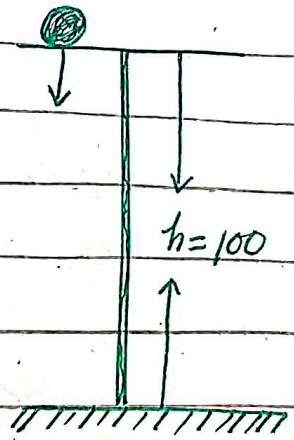
Since:

$v^2 - u^2 = 2gh$

$v^2 - 0 = 2(32)(100)$

$v^2 = 6400$

$\Rightarrow v = 80 \text{ ft/sec}$



Now:

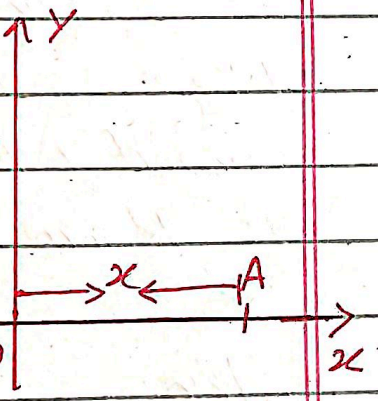
$v = gt + u$

$80 = (32)t + 0$

$\Rightarrow t = \frac{80}{32} = 2.5 \text{ sec}$

"The Simplist motion in nature along the stright line is called Rectilinear motion.

Let 'O' be a fixed pt and 'x' be the distance of particle from 'O' to any point 'A' Cover in time 't'



Then the magnitude of velocity 'v' and Acc are:-

$v = \frac{dx}{dt}$ —① $\quad a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$ —②

Also, $a = \frac{dv}{dt} = \frac{dv}{dx} \cdot \frac{dx}{dt} = v\frac{dv}{dx}$ —③

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Velocity dependent Acc:-

If Acceleration of a Particle depends upon velocity, Then such an acc is called velocity dependent Acc.

Consider:-

$$v\frac{dv}{dx} = a(v)$$

$$dx = \frac{v}{a(v)}dv$$

Integrate $$\int dx = \int \frac{v\,dv}{a(v)}$$

$$x = \int \frac{v}{a(v)}dv + C$$

Where Constant 'C' can be determined when velocity of the particle is Known for some value of 'x'.

$$Acc = \frac{dv}{dt}$$

$$= \frac{dv}{dt}\cdot\frac{dx}{dx}$$

$$= v\frac{dv}{dx}$$

$$Acc = v\frac{dv}{dx}$$

Also:- $$\frac{dv}{dt} = a(v)$$

$$dt = \frac{dv}{a(v)} \Rightarrow \int dt = \int \frac{dv}{a(v)}$$

$$t = \int \frac{dv}{a(v)} + D$$

Where Constant 'D' can be determined if the velocity at some instant is

Example:- A Particle move in stright line with an aueleration Kv^3. if its initial velocity is 'u' find the velocity and time spent when the particle has travelled a distance "x"

Solution:- $a(v) = Kv^3$, $v=?$, $t=?$

Bourndary Conditions:- at,

$t=0$, $x=0$, $v=u$

$$v\frac{dv}{dx} = Kv^3$$

$$\Rightarrow v\,dv = Kv^3\,dx$$

$$v^{-2}\,dv = K\,dx$$

$$\int v^{-2}\,dv = K\int dx$$

$$\Rightarrow \frac{v^{-1}}{-1} = Kx + A$$

$$-\frac{1}{v} = Kx + A \quad \text{---(1)}$$

Put, $x=0$, $v=u$

$$\boxed{A = -\frac{1}{u}}$$

Hence Eq (1) become

$$-\frac{1}{v} = Kx - \frac{1}{u}$$

OR

$$\frac{dv}{dt} = Kv^3$$

$$\Rightarrow dv = Kv^3\,dt$$

$$\Rightarrow v^{-3}\,dv = K\,dt$$

$$\Rightarrow \frac{v^{-2}}{-2} = K\cdot t + A$$

$$\Rightarrow -\frac{1}{2v^2} = Kt + A$$

$$\Rightarrow v^2 = -\frac{1}{2Kt} + A$$

No, possible the calculate the value of v or t

$$\Rightarrow -\frac{1}{v} = \frac{Kux-1}{u} \Rightarrow \boxed{v = \frac{u}{1-Kux}}$$

$$\boxed{\because v = \frac{dx}{dt}}$$

$$\frac{dx}{dt} = \frac{u}{1-Kux} \Rightarrow dx = \frac{u}{1-Kux}dt$$

$$\Rightarrow (1-Kux)dx = u\,dt$$

Integrate Both sides

$$\Rightarrow \int dx - Ku\int x\,dx = u\int dt$$

$$x - K\cdot u\frac{x^2}{2} = ut + B - (2)$$

Put $x=0, t=0$ in Eq (2)

$$\boxed{B=0}$$

Hence Equation (2) becomes

$$x - K\cdot u\cdot\frac{x^2}{2} = ut$$

$$\Rightarrow t = \frac{1}{u}\left(x - Ku\frac{x^2}{2}\right)$$

$$\Rightarrow \boxed{t = \frac{x}{u}\left(1 - \frac{Kux}{2}\right)} \text{ Ans}$$

(1) Q :- A Particle start with velocity 'u' and move in stright line. if it sufferis a ~~rect~~ retardation equal to the square of the

velocity. Find the distance travelled by the particle in a time 't'

Solution: $a(v) = -v^2$, $x = ?$

Boundry Conditions: $t = 0 \vee$
$x = 0 \vee$
$v = u \vee$

$\left(a(v) = \frac{dv}{dt} \;,\; a(v) = v\frac{dv}{dx} \right)$

$\Rightarrow \frac{dv}{dt} = -v^2$

$\Rightarrow -dv = v^2 dt$

$-\frac{dv}{v^2} = dt$

$-v^{-2} dv = dt$

Integrate Both sides

$-\frac{v^{-1}}{-1} = t + A$ ①

$\Rightarrow \frac{1}{v} = t + A$

Put $t = 0, v = u$

$\boxed{A = \frac{1}{u}}$

Hence Eq ① becomes

$\Rightarrow \frac{1}{v} = t + \frac{1}{u}$

OR

$\Rightarrow v\frac{dv}{dx} = -v^2$

$\Rightarrow -v\,dv = v^2 dx$

$\frac{-v}{v^2} dv = dx$

$-v^{-1} dv = dx$

$-\frac{dv}{v} = dx$

Integrates Both sides

$-\int \frac{dv}{v} = \int dx$

$x = -\ln v + A$ ①

Put. $x = 0, v = u$

$-\ln(u) + A = 0$

$\boxed{A = \ln u}$

$x = \ln u - \ln v$

$$\Rightarrow \frac{1}{v} = \frac{ut+1}{u}$$

$$\Rightarrow \boxed{v = \frac{u}{ut+1}}$$

$$\Rightarrow \frac{dx}{dt} = \frac{u}{ut+1}$$

$$\Rightarrow dx = \left(\frac{u}{ut+1}\right)dt$$

$$\Rightarrow \int dx = \int \left(\frac{u}{ut+1}\right)dt$$

$$x = \ln(ut+1) + B - (2)$$

Put $t=0, x=0$ In Eq (2)

$$\boxed{B = 0}$$

Hence Equation (2) Becomes

$$\boxed{x = \ln(ut+1)}$$

$$\therefore \ln e = 1$$

$$x = \ln\left(\frac{u}{v}\right)$$

Taking anti-log

$$e^x = e^{\ln\left(\frac{u}{v}\right)}$$

$$\frac{u}{v} = e^x$$

$$u = ve^x$$

$$\Rightarrow v = \frac{u}{e^x}$$

$$\Rightarrow \frac{dx}{dt} = \frac{u}{e^x}$$

$$e^x dx = u\,dt$$

$$\Rightarrow \int e^x dx = u \int dt$$

$$e^x = ut + B - (2)$$

Put $x=0, t=0$

$$\boxed{B = 1}$$

$$e^x = ut + 1$$

$$\ln e^x = \ln(ut+1)$$

$$x \ln e = \ln(ut+1)$$

$$\boxed{x = \ln(ut+1)}$$

(2) :- A Particle moving with a stright line starts with vel: 'u' and Acc: v^3 where 'v' is the velocity at time 't'. Find velocity

and time as function of distance travelled by particle.

Solution:- $a(v) = v^3$, $v = ?$, $t = ?$

Boundry Conditions: $t=0, x=0, v=u$

$$\frac{v\,dv}{dx} = v^3$$

$$\Rightarrow v^{-2}\,dv = dx$$

$$\Rightarrow \int v^{-2}\,dv = \int dx$$

$$\Rightarrow \frac{v^{-1}}{-1} = x + A \quad \text{(1)}$$

$$\Rightarrow -\frac{1}{v} = x + A \quad \text{(1)}$$

Put $x=0, v=u$

Put in Eq (1)

$$\boxed{A = -\frac{1}{u}}$$

Hence Eq (1) Becomes

$$-\frac{1}{v} = x - \frac{1}{u}$$

$$\Rightarrow -\frac{1}{v} = \frac{ux-1}{u}$$

$$\Rightarrow \boxed{v = \frac{u}{1-ux}}$$

OR

$$\frac{dv}{dt} = v^3$$

$$\Rightarrow v^{-3}\,dv = dt$$

$$\Rightarrow \int v^{-3}\,dv = \int dt$$

$$\Rightarrow \frac{v^{-2}}{-2} = t + A \quad \text{(1)}$$

$t=0, v=u$ in (1)

$$\boxed{A = \frac{u^{-2}}{-2}}$$

$$-\frac{v^{-2}}{2} = t - \frac{u^{-2}}{2}$$

$$\Rightarrow t = \frac{1}{2u^2} - \frac{1}{2v^2}$$

Here, 't' is in term of (v) (not possible)

$$\Rightarrow \frac{dx}{dt} = \frac{u}{1-ux}$$

$$\Rightarrow (1-ux)\,dx = u\,dt$$

$$\Rightarrow \int (1-ux)\,dx = u\int dt$$

$$\Rightarrow x - u \cdot \frac{x^2}{2} = ut + B \quad \text{—②}$$

Put $t = 0$, $x = 0$ in eq ②

$$\boxed{B = 0}$$

Hence Eq ② Becomes

$$x - u \cdot \frac{x^2}{2} = ut$$

$$\Rightarrow ut = x - u \cdot \frac{x^2}{2}$$

$$\Rightarrow t = \frac{x}{u} - \frac{x^2}{2}$$

$$\Rightarrow t = \frac{2x - u \cdot x^2}{2u}$$

$$\Rightarrow t = \frac{x}{2u}(2 - ux) \quad \underline{\text{Ans}}$$